TAREK SHEREEN DARWISH

Software Engineer: Systems, Rust & Open Source

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SUMMARY

Systems-minded software engineer finishing a B.Sc. in Intelligent Systems Engineering. Designs and builds low-level Rust and C projects: a Windows game recorder with a GPU-to-disk capture path and an allocation-free DVR buffer, and ESP32 firmware for a health band. Merged fix in librespot (#1786) and six merged PRs to Sonora (Spotify Connect, live Liked Songs, Jams, right-to-left text). Winner of the Youth Leaders Foundation (YLF) Hackathon 2026 and 2nd place at Banha, both with Healthica. Looking for a junior or graduate software engineering role, available from February 2027.

TECHNICAL SKILLS

Languages: Rust, C, C++, Python, SQL; also TypeScript/JavaScript, Dart, Kotlin, Java

Systems: Direct3D 11, Media Foundation, WASAPI, Win32 API, BLE (GATT), ESP32 firmware (C/Arduino), llama.cpp

Linux & tools: Linux (Arch), bash and PowerShell scripting, WSL2, Git, Docker, CI/CD, OCI CLI

Cloud & ML: Azure, AWS, Cloudflare, Supabase, TensorFlow Lite, PyTorch, scikit-learn

OPEN SOURCE

- librespot (Rust, Spotify client library): merged PR #1786, github.com/librespot-org/librespot/pull/1786. Liked Songs updates arrive as plain JSON but were decoded as base64 and dropped; the fix passes them on as JSON. Small change (+6 -3 lines).
- Sonora (Rust music app, 2.5k stars), 6 merged PRs (Oct to Nov 2026): Spotify Connect device #895 (+2,809 -93, 51 files), live Liked Songs #905, Spotify Jams #934, Jam leave and queue fix #1170, right-to-left text #880 with sonorahq/gpui #18, autoplay default #935. Open: running downloads in title bar #1038.
- Also open: librespot #1804 (docs for Spotify Jam endpoints), MeasureIt_ARCH #309 (Python, Blender add-on).

PROJECTS

ZeroClip: Windows Game Clipper · *Creator and sole developer · pre-alpha · Rust, Direct3D 11, Media Foundation, WASAPI*

- GPU screen capture feeding hardware H.264/HEVC encoders with frames kept in VRAM, WASAPI audio, an allocation-free, GOP-aware DVR buffer, and a fragmented MP4 writer that saves clips without re-encoding.

Healthica: Smart Band Health Monitor · *Winner, YLF Hackathon 2026 (team of 5) · 2nd place, Banha Hackathon · C, ESP32, BLE, TFLite*

- Worked on the ESP32 firmware: reads a MAX30105 PPG sensor and a GSR input, detects beats, computes HRV (RMSSD, SDNN, pNN50) and sends a payload of about 250 bytes over BLE GATT notifications, with a WiFi HTTP fallback.
- Split the work to keep the band light: HRV and PPG features on the ESP32, inference on the phone with INT8 TensorFlow Lite glucose models of 8 to 23 KB. Also built the Flutter app and web dashboard.

safi: Local Agentic LLM Backend · *Python, llama.cpp, PowerShell · in progress since Oct 2026*

- Planner and executor model pair (Qwen3.8-27B, Qwen3.6-35B-A3B) on llama.cpp, 16 GB VRAM laptop. Built an onboarding probe that checks every layer is on the GPU and measures VRAM, prefill and decode speed, plus a 10 minute thermal soak.
- Also built a tool registry, a CPU embedder and PowerShell/shell router supervisors, with a pytest suite.

OpenCrosshair: Windows Crosshair Overlay · *Open source (MIT) · github.com/Tu2525/OpenCrosshair · Rust, Win32 API, egui*

- Click-through overlay with per-game presets. No DLL injection by design; tradeoff: no exclusive fullscreen.

EXPERIENCE

Generative AI Engineer, Digital Egypt Pioneers Initiative — *Cairo, Egypt (Hybrid) · Nov 2025 – Jul 2026*

- Built and deployed (Docker, Hugging Face Spaces) an explainable fraud-detection service: a 310-feature classifier behind FastAPI with Gemini-written explanations from SHAP.

Backend & AI Engineer Intern, Aispeak — *Madrid, Spain (Remote) · Jul 2025 – Jan 2026*

- Built a backend pipeline routing user messages to a LLaMA-70B model on Azure, plus Supabase real-time messaging.

EDUCATION & CERTIFICATIONS

B.Sc., Intelligent Systems Engineering, Helwan National University — *Helwan, Egypt · 2022 – Jan 2027 (expected)*

DEPI Generative AI and Mobile Tracks | AWS Cloud Practitioner | Huawei HCIP-AI | Huawei ICT Associate | Stanford ML Specialization